

PORTLAND INTL, OR, USA

WMO: 726980

Lat: 45.596N

Lon: 122.609W

Elev: 6

StdP: 101.26

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24229

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-3.4	-1.4	-11.7	1.4	-0.2	-8.5	1.8	1.5	13.8	3.1	12.3	4.1	4.2	120	0.509

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.0	33.2	19.6	30.8	19.1	28.8	18.3	20.8	30.6	19.9	29.2	19.0	27.4	4.5	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	12.3	23.8	16.3	11.6	22.9	15.6	11.1	22.0	59.7	30.7	56.5	29.1	53.6	27.4	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	8.7	7.7	DB	-6.0	37.4	2.9	1.9	-8.0	38.7	-9.7	39.8	-11.3	40.9	-13.4	42.3	(4)
(5)			WB	-6.9	22.5	2.8	1.0	-8.9	23.2	-10.5	23.7	-12.1	24.3	-14.1	25.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.7	5.4	6.5	8.9	11.4	15.0	17.7	21.1	21.3	18.4	12.9	8.4	5.1	(6)	
	DBStd	6.48	3.19	2.98	2.71	2.99	3.38	3.22	2.95	2.66	2.96	2.76	3.19	3.25	(7)	
	HDD10.0	546	146	102	54	17	2	0	0	0	0	6	66	153	(8)	
	HDD18.3	2322	402	332	293	211	114	49	7	4	36	168	298	409	(9)	
	CDD10.0	1536	2	3	19	58	156	232	344	351	252	98	18	3	(10)	
	CDD18.3	269	0	0	0	1	11	30	93	96	37	1	0	0	(11)	
	CDH23.3	2559	0	0	1	26	152	321	870	822	355	13	0	0	(12)	
	CDH26.7	970	0	0	0	4	44	118	356	328	118	1	0	0	(13)	
(14) Wind	WSAvg	3.2	4.0	3.7	3.4	3.1	2.9	3.0	3.1	2.9	2.7	2.8	3.5	4.1	(14)	
(15) Precipitation	PrecAvg	1177	172	115	124	93	74	50	14	19	42	109	177	189	(15)	
	PrecMax	1854	335	288	215	191	148	118	59	83	151	242	401	397	(16)	
	PrecMin	902	62	10	47	39	4	9	0	0	0	22	61	70	(17)	
	PrecStd	212	63	57	43	37	38	24	12	17	30	54	78	80	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.4	15.8	21.2	26.7	30.7	34.4	36.6	35.7	32.9	25.1	18.0	14.7	(19)	
		MCWB	12.4	10.5	13.0	15.6	17.3	20.0	21.1	20.1	18.7	15.4	14.6	12.8	(20)	
	2%	DB	12.4	13.9	18.1	22.7	27.5	30.0	33.2	32.7	30.0	22.0	15.7	12.3	(21)	
		MCWB	10.1	9.8	11.2	14.0	16.5	18.4	20.1	19.5	17.6	14.2	12.8	10.6	(22)	
	5%	DB	11.0	12.4	15.6	19.5	24.4	26.9	30.5	30.3	27.5	19.8	14.3	10.7	(23)	
		MCWB	9.0	8.9	10.2	12.1	15.4	17.3	19.6	19.3	17.3	13.7	11.9	9.1	(24)	
	10%	DB	9.7	10.9	13.7	16.9	21.6	24.2	28.1	28.0	24.9	17.6	12.9	9.4	(25)	
		MCWB	7.9	7.9	9.5	11.2	14.3	16.2	18.6	18.5	16.6	13.4	10.8	7.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	12.7	14.0	16.5	18.9	21.0	22.6	21.5	20.4	17.9	15.5	13.1	(27)	
		MCDB	14.2	14.3	19.6	25.0	28.1	31.7	33.7	32.3	29.4	22.1	17.0	14.5	(28)	
	2%	WB	10.6	10.8	12.3	14.5	17.4	19.1	21.0	20.5	19.1	16.0	13.5	10.8	(29)	
		MCDB	11.8	12.7	15.9	21.3	25.2	28.4	30.8	30.4	27.1	19.6	15.1	12.0	(30)	
	5%	WB	9.3	9.5	11.1	13.0	16.1	17.8	20.0	19.7	18.1	15.0	12.3	9.4	(31)	
		MCDB	10.7	11.7	14.3	18.1	22.9	25.5	29.3	28.9	25.1	17.9	13.8	10.6	(32)	
	10%	WB	8.1	8.4	10.2	11.8	14.9	16.7	18.9	18.8	17.2	14.0	11.2	8.0	(33)	
		MCDB	9.5	10.5	13.2	15.7	20.6	23.3	27.4	27.0	23.2	16.6	12.6	9.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	7.1	8.3	9.1	9.9	10.3	12.2	12.0	11.2	8.5	6.1	5.0	(35)	
		MCDBR	6.7	9.3	12.3	14.7	15.8	15.7	16.4	16.3	16.3	12.9	7.2	6.0	(36)	
	5% WB	MCWBR	4.9	5.6	6.6	7.0	6.9	6.4	6.3	5.9	6.1	6.1	4.6	4.8	(37)	
		MCWBR	6.1	7.7	10.0	12.8	13.9	14.2	15.3	14.8	13.8	9.1	5.8	5.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.321	0.317	0.330	0.362	0.377	0.362	0.350	0.357	0.344	0.334	0.322	0.326	(40)	
		taud	2.531	2.552	2.524	2.400	2.375	2.432	2.490	2.486	2.505	2.543	2.540	2.477	(41)	
	Ebn at Noon		792	873	906	898	891	904	911	892	878	837	783	737	(42)	
		Edh at Noon	66	77	90	111	118	112	104	101	91	76	64	63	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.16	2.00	2.90	4.23	5.39	5.95	6.78	5.79	4.32	2.50	1.35	0.94	(44)	
	RadStd		0.19	0.28	0.38	0.40	0.52	0.56	0.48	0.39	0.38	0.31	0.18	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+39
(47) Regional (32 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	+26

Nomenclature: See separate page